KARSHI STATE UNIVERSITY



SDG PROGRESS REPORT
2024–2025

Progress towards the United Nations Sustainable Development Goals

Karshi, Uzbekistan
2026

Karshi State University (KSU) is committed to integrating sustainable development into education, research, campus operations, social responsibility, community engagement and institutional governance. This SDG Progress Report 2025–2026 consolidates documented activities and results that contribute to the United Nations Sustainable Development Goals (SDGs). The underlying Sustainability Report states that it covers activities, data and progress achieved during the 2025–2026 academic year and is coordinated through the Department of International Rankings and Sustainable Development together with a cross-functional Sustainability Council. The University also established a permanent Sustainability Committee in April 2025 to coordinate SDG implementation, monitor indicators and prepare annual sustainability reports.

This publication reorganizes documented evidence by SDG so that the University's contribution to each goal can be understood directly. Where the available institutional evidence is stronger for some SDGs than others, the report does not invent figures; instead, it identifies the evidence currently documented and notes where additional SDG-specific institutional data would strengthen future reporting.

Reporting period

2024–2025 academic year, with selected multi-year indicators where the source report explicitly uses a longer reporting period (e.g., 2023–2025 research and budget data). University sustainability policy and institutional governance information contained in the Sustainability Report.

University news and activity records supplied for inclusion in the SDG evidence base, including agricultural research, staff social support, professional training, and international cooperation.

Institutional framework for SDG implementation

KSU coordinates sustainability through the Department of International Rankings and Sustainable development and a cross-functional Sustainability Council involving faculties, student organizations and administrative units. The Sustainability Committee was formally established by the Academic Council on 3 April 2025 as a permanent body responsible for coordinating and monitoring sustainability policies, activities and performance indicators.

-coordinate university initiatives related to sustainable development and SDG implementation.

-prepare annual sustainability reports.

-monitor energy, water and waste indicators with relevant departments.

-organize awareness campaigns, student engagement and sustainability-related research.
-support transparent data management and international collaboration.

SDG 1- No poverty

End poverty in all its forms everywhere

Karshi State University (KSU) implements a range of social sustainability measures aimed at reducing financial vulnerability and supporting the well-being of students, employees and socially vulnerable groups. These measures include staff welfare programmes, student support, social assistance, financial incentives and the development of an inclusive campus environment.

In 2025, more than USD 2.4 million was allocated to social support and incentive programmes benefiting employees, students and socially vulnerable groups. This support forms part of the University's broader sustainability investment framework and is intended to improve access to essential services, education and institutional support for individuals facing financial or social challenges.

KSU also provides an inclusive campus environment for people with disabilities, special needs and employees requiring maternity-related facilities, supporting equal access to education, employment and university services.

KSU provides direct financial assistance to students experiencing financial hardship and difficult family circumstances. This support is formally documented through university administrative orders on material assistance and financial incentives.

For example, Rector's Order 2025 provided direct financial assistance to students facing severe social and family circumstances. The order specifically identifies students who had lost one or both parents, were living under the care of relatives, had lost their breadwinner, or were experiencing difficult family conditions. Depending on the individual circumstances, students were granted financial assistance equivalent to three, four or five basic scholarship amounts**.** Further support was provided through Rector's Order 2025**,** under which students requiring social assistance and students actively participating in academic and community activities received financial incentives equivalent to one basic scholarship. The documented reasons for support included being socially vulnerable, having lost a breadwinner, and being socially active**.**

In addition, Rector's Order 2025 provided financial incentives to students requiring social support in connection with the New Year period. The order confirms that financial support was provided from the University's Student Financial Incentive Fund.

These documented measures demonstrate that KSU's approach to poverty reduction includes targeted financial assistance for students facing economic vulnerability, loss of a breadwinner and difficult family circumstances, alongside broader institutional social-support programs.

Documented indicators and results

- social support allocation: More than USD 2.4 million allocated in 2025.
- direct student financial assistance: Documented through Rector's Orders
- individual financial assistance: Support equivalent to 3–5 basic scholarship amounts was provided in documented cases of severe social and family hardship.
- support for socially vulnerable students: Assistance included students who had lost a breadwinner and students experiencing difficult social or family circumstances.
- food affordability initiative: An affordable food products fair was organized on December 2025**.**
- direct staff support: Free meat was distributed to employees in need following the slaughter of three cattle.
- inclusive campus support: Facilities and services are provided to support people with disabilities, special needs and maternity-related needs.

Karshi State University contributes to economic resilience and poverty reduction by supporting graduates' transition into employment, including employment in emerging green sectors. During the 2024/2025 academic year, 6,052 students graduated from KSU, of whom 557 obtained green jobs, representing approximately 9.2% of all graduates. The development of employment opportunities provides graduates with pathways to income generation and economic independence, while green employment also connects graduate employability with the transition towards a more sustainable economy. KSU therefore considers graduate employment and the development of green-job opportunities an important component of its contribution to SDG 1, particularly through strengthening young people's access to employment and livelihoods.

KSU supports sustainable livelihoods and food security through research and practical activities focused on sustainable agricultural production. One of the University's research projects, focused on the cultivation technology of edible and medicinal mushrooms in the Kashkadarya region, received 981.92 million UZS in research funding. The project addresses sustainable agriculture, food security and

biodiversity preservation and contributes to the development of locally relevant agricultural knowledge and production technologies. By supporting agricultural innovation and food production in the region, KSU contributes to strengthening food security and creating knowledge that can support sustainable livelihoods, particularly in an economically important agricultural context such as Kashkadarya.

KSU promotes student innovation and economic empowerment through financial incentives for students developing environmentally and socially oriented solutions. In the Eko-Innovation–2025 competition, students were encouraged to develop innovative solutions related to environmental sustainability and the green economy, with financial awards of 6 million UZS for first place, 4 million UZS for second place and 2 million UZS for third place. Such initiatives provide students with direct financial incentives while encouraging them to develop innovative ideas and entrepreneurial capabilities. Through these activities, KSU supports young people's participation in innovation and creates opportunities for students to develop skills and ideas that may contribute to future employment, entrepreneurship and income generation.

KSU contributes to long-term economic empowerment by providing education and professional training in fields directly connected with employment, financial management and economic activity. The University's academic programs include Economics, Finance and Financial Technologies, Banking, Accounting, Marketing, Tourism and Hospitality, together with sustainability-oriented subjects such as Green Economy, Energy Management and the Organization and Management of Greenhouse Economy. These programs equip students with professional, financial and economic knowledge that can support their future participation in the labour market and broader economy. By developing human capital and employment-related competencies, KSU contributes to improving students' capacity to secure sustainable livelihoods and participate in economic development.

KSU promotes social inclusion by reducing barriers that may prevent vulnerable groups from fully participating in higher education and university life. The University has implemented accessibility measures including wheelchair ramps, accessible toilets, disabled parking spaces, tactile paving, Braille signage and accessibility lifts, supporting students and employees with disabilities and special needs. In addition, KSU provides free psychological consultations, seminars and workshops through its psychological support services. These measures help ensure that access to education and essential university services is not unnecessarily restricted by disability or social circumstances. By creating a more inclusive institutional environment and providing support services without additional financial barriers, KSU contributes to reducing vulnerability and promoting equal participation in education and employment.

SDG 2 – Zero hunger

End hunger, achieve food security and improved nutrition and promote sustainable agriculture

Karshi State University contributes to SDG 2 – Zero Hunger through sustainable agriculture, food-security research, conservation of genetic resources for food and agriculture, sustainable food production, biotechnology, and agricultural education. The University maintains a main campus greenhouse where tomatoes and cucumbers are cultivated under controlled environmental conditions, with ICT-based systems used to monitor temperature, humidity and irrigation and support sustainable horticultural practices. KSU also operates a mushroom greenhouse that conserves and cultivates more than 10 varieties of edible mushrooms and an equivalent number of medicinal mushrooms; smart monitoring systems regulate light, humidity and temperature, supporting scientific research, food sustainability and sustainable agriculture. In addition, the University has an innovative potato field conservation project that uses ICT-enabled soil and crop monitoring to improve the sustainability and genetic diversity of potato cultivation.

KSU also supports food security through dedicated agricultural research. The project "Development of a cultivation technology for edible and medicinal mushrooms distributed in the Kashkadarya region under artificial nutrient conditions" was implemented in 2024–2025 with funding of 981,920,000 UZS, with its stated focus on sustainable agriculture, food security and biodiversity preservation. The report explicitly identifies this research as contributing to SDG 2. The University also organized an international event on "The relevance of using innovative technologies in the cultivation of organic agricultural products in different soil and climate conditions," which involved 150 participants, including 10 foreign scholars, providing a platform for knowledge exchange on innovative and sustainable agricultural production.

The University's contribution to sustainable agriculture is further supported through its academic programs and research capacity. The Geography and Agronomy area includes programs such as Plant Protection and Quarantine, Agrochemistry and Soil Science, Soil Science, and Ecology and Environmental Protection, while 51 lecturers and researchers are associated with the Geography and Agronomy faculty. KSU also offers sustainability-related courses including Biochemistry of Agricultural Products, Agrochemical Services in Agriculture, and Organization and Management of Greenhouse Economy, providing students with knowledge relevant to agricultural production, soil management, food-related science and sustainable land use.

Food-resource efficiency is another relevant component of KSU's SDG 2 contribution. In 2025, 209.3 tonnes of food waste were generated mainly from campus cafeterias and dormitories, representing a reduction of 108.1 tonnes compared with the previous year. The University operates a segregated organic-waste collection and treatment system, and 18.7 tonnes of food waste were processed and reused as animal feed through a granulation process. In addition, 18 tonnes of leaf waste were converted into nutrient-rich biomass fertilizer using California red worms and used to improve soil quality in university green areas. These practices demonstrate the University's efforts to reduce food-resource losses and promote circular approaches to agricultural and organic resources.

KSU's Environmental Sustainability Policy includes sustainable procurement and consumption measures and explicitly commits the University to promoting sustainable food and catering services on campus. The policy also supports sustainable water management, including drip irrigation and wastewater recycling where feasible, which is relevant to the sustainable management of agricultural and food-production resources.

University scientists from the Agrochemistry and Ecology Department developed and tested locally adapted sweet potato varieties, including Sochakinur, Toyloqi, Filial and New Samarkand. Field trials in Karshi District reported yields of 50–60 tonnes/ha for the primary crop and 42–45 tonnes/ha for the secondary crop, with individual plant yields of 800–1,000 g or more. These results provide particularly strong quantitative evidence of KSU's contribution to sustainable food production and agricultural innovation.

SDG 3 – Good health and well-being

Ensure healthy lives and promote well-being for all at all ages

Karshi State University (KSU) contributes to SDG 3 – Good Health and Well-being through the provision of healthcare services, preventive health programs, psychological support, medical education, sports and physical activity, safe drinking water, emergency preparedness, accessible infrastructure, and a safe and healthy campus environment. The university's Sustainability Report documents a range of facilities and institutional measures aimed at protecting and improving the health and well-being of students, academic staff, administrative staff, and visitors.

KSU provides accessible medical care and first-aid facilities across the university. Medical treatment rooms are available in different academic buildings to provide immediate healthcare assistance to members of the university community. The report

specifically identifies a medical room in the Medical Faculty study block, a medical room in the Physics and Mathematics Faculties study block, a medical treatment room in the Foreign Language Faculty study block, the University Psychologist's Office, and the Karshi State University Medical Center. Medical Faculty staff and students also contribute to the university community by providing first-aid services and professional medical consultations when required.

The university also places emphasis on mental health and psychological well-being. Psychological support is available to help students and staff manage stress and maintain their overall well-being. The University Psychologist's Office provides free consultations, seminars, and workshops, creating an institutional mechanism for addressing psychological needs and promoting a healthy academic environment.

KSU supports preventive healthcare and health promotion through collaboration with local healthcare providers. Students and staff have access to comprehensive medical services, including preventive care and specialized treatment. The university organizes health-awareness programs, vaccination campaigns, and regular medical check-ups aimed at promoting healthy lifestyles within the academic community. These activities demonstrate the university's focus not only on treatment but also on disease prevention and health awareness.

Health and well-being are also supported through KSU's medical education and research capacity. The university has 109 lecturers and researchers in the field of Medicine, whose work is described as contributing to the improvement of public health and sustainable medical practices. In addition, the university has 37 professionals in Sport, contributing to health, physical education, and sustainable well-being. This academic and professional capacity enables KSU to contribute to health-related education, research, physical education, and public-health development.

The university promotes physical activity and healthy lifestyles through sports infrastructure and activities. The main campus provides indoor and outdoor sports facilities, while the Philology Faculty Campus includes a football field and a gym. The university also supports its football team by providing a 36-seat bus for travel to games. These facilities and activities provide opportunities for students and staff to participate in physical exercise and organized sport, contributing to physical health and overall well-being.

Access to safe and high-quality drinking water is another important contribution to health and well-being. KSU operates an ASMOS Water Recycling and Purification System, installed in 2023. Water from 15 artesian wells is purified through multi-stage filtration using ASMOS technology. The purified water is supplied to academic buildings, laboratories, administrative offices, dormitories, and campus cafeterias,

ensuring that students and staff have access to safe, high-quality drinking water. The system has a total processing capacity of 300 m^3 per hour.

The university also maintains sanitary and hygiene facilities throughout its campuses. Water-saving devices, including low-flow taps and dual-flush toilets, are installed in restrooms and washrooms, while regular monitoring and maintenance are conducted to ensure their effective operation. Accessible toilets are provided in the main buildings, including faculties, dormitories, the Rectorate, and the library. These facilities contribute to maintaining appropriate sanitary conditions and supporting the health, dignity, and comfort of students, staff, and visitors.

KSU has also established measures to protect the university community from accidents and emergencies. The campus is equipped with CCTV surveillance systems, an emergency hotline, alarm systems, and trained and certified security personnel. Fire safety infrastructure includes fire extinguishers, hydrants, and alarm systems, which are regularly inspected and maintained. The university conducts regular safety drills and awareness programmes to prepare students and staff for emergency situations. The university's response time to accidents, crimes, fires, or natural disasters is less than five minutes, supporting rapid emergency response and risk reduction.

The university promotes inclusive health and well-being by providing accessible infrastructure for people with disabilities, special needs, and maternity care. KSU provides disabled parking spaces, accessible toilets, wheelchair ramps, tactile paving, Braille signboards, and accessibility lifts. These facilities are designed to ensure equal access to education, mobility, comfort, and university services for students, staff, and visitors with different accessibility needs.

KSU's transportation policy also contributes indirectly to a healthier campus environment. Private cars and motorcycles are not permitted to operate within the university's 319,107 m^2 campus, while the university operates seven electric motorcycles and two 10-seat electric golf shuttles. Bicycles are widely used by students and staff, and dedicated bicycle paths connect key campus facilities. The restriction of conventional motor vehicles and promotion of low-emission and bicycle-based mobility contributes to a cleaner, pedestrian-friendly campus environment and supports active mobility.

The university's physical environment further supports well-being through extensive green and landscaped spaces. The main campus covers 285,356.83 m^2, representing 89.4% of the university's total area, and includes landscaped areas and green spaces. The overall university area is 319,107 m^2. These spaces form an important part of the university's sustainable campus environment and provide an appropriate setting for recreation, movement, and everyday university activities.

Karshi State University promotes an active and healthy lifestyle among students residing in university dormitories through daily morning physical exercise activities (Badantarbiya mashg'uloti). These organized outdoor exercise sessions are conducted regularly on the university campus and involve dormitory students in collective physical activities. The initiative encourages regular exercise, improves physical fitness, supports healthy lifestyle habits, and contributes to the physical and psychological well-being of students. As a regular university-wide health promotion activity, the programme provides direct evidence of KSU's contribution to SDG 3 – Good Health and Well-being.

SDG 4 – Quality education

Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

Karshi State University contributes to SDG 4 by providing broad and multidisciplinary educational opportunities across humanities, natural sciences, engineering, technology, medicine, social sciences, pedagogy, sport, and other fields. The university offered 860 courses in 2023, 873 courses in 2024, and 846 courses in 2025, demonstrating a stable and diverse academic portfolio. The curriculum is continuously updated in accordance with national higher education requirements and international academic quality benchmarks, with the aim of providing students with relevant and globally competitive knowledge.

Quality education is strengthened through the integration of sustainability and interdisciplinary knowledge into academic programs. KSU provides 169 courses with sustainability-related content, covering environmental protection, renewable energy, biotechnology, social responsibility, agriculture, health, ecology, and sustainable development. These courses are offered across Engineering and Technology, Chemistry and Biology, Geography and Earth Sciences, Physics, Economics, Medicine and Health, and Pedagogy. Examples include Alternative Energy and Nature Protection, Energy Efficient Technologies, Bioengineering, Eco biotechnology, Chemical Ecology and Environmental Protection, Applied Ecology, Environmental Impact Assessment, Green Economy, Energy Management, Ecological Physiology, Fundamentals of Health Preservation, Methods of Teaching Ecological Sciences, and Developmental Psychology.

KSU provides students with modern digital learning opportunities through the extensive use of ICT in classrooms and lecture facilities. Educational spaces are equipped with computers, smart TVs, projectors, interactive panels, and other digital technologies that support multimedia presentations, online resources, and hybrid

teaching. The university also uses Zoom and other digital platforms for lectures, academic meetings, and international collaboration. Language and translation training facilities are equipped with voice amplifiers and digital recording equipment to support communication practice, simultaneous interpretation, and performance assessment.

The university's educational infrastructure includes 1,305 computers, 128 video projectors, 44 interactive panels, 10 information kiosks, 103 smart televisions, 152 Wi-Fi routers, 4 E-MINBAR systems, 16 computer laboratories, 140 computers in the Information Resource Centre, and 96 computers in the Student Dormitory. These facilities expand students' access to digital learning resources and support technology-enhanced teaching and independent learning.

KSU also provides an environment that combines formal education with research and practical learning. Academic staff are actively engaged in research, while students participate in sustainability projects, scientific activities, competitions, environmental campaigns, and innovation initiatives. During 2024–2025, students participated in more than 30 sustainability-related educational and practical activities, including scientific dialogues, intellectual competitions, environmental awareness programs, youth camps, and student innovation projects. Undergraduate, graduate, and doctoral students aged 18–35 also participated in the Eko-Innovation–2025 competition, developing solutions related to environmental protection, recycling, climate change mitigation, renewable energy, and sustainable technologies.

Student development extends beyond classroom instruction through cultural, educational, intellectual, and social activities. During the 2024–2025 academic year, more than 100 cultural and educational events were organized, including Student Spring, Language – the Mirror of the Nation, Book Reading Week, National Values Festival, Spiritual Night, and Theatre Week. Concerts, competitions, public discussions, theatre performances, exhibitions, and other activities were organized in faculties, dormitories, and on campus, providing students with opportunities for personal development, cultural participation, communication, creativity, and social interaction.

The university also promotes research-based education and knowledge creation. In 2024-2025, KSU produced 745 scholarly publications indexed in Google Scholar and 267 publications indexed in Scopus. Research activities cover renewable energy, environmental protection, sustainable agriculture, water-resource management, climate change, green economy, sustainable business development, education for sustainable development, social responsibility, and ecological linguistics. The university's research activities create opportunities for students and academic staff to engage with contemporary scientific and societal challenges.

Research and innovation are supported by dedicated funding. Sustainability-related research funding amounted to 1,403,660,563 UZS in 2023, 1,578,764,237 UZS in 2024, and 1,529,581,000 UZS in 2025, with an average annual amount of approximately 1,504,001,933 UZS. The supported research portfolio includes projects in mushroom cultivation, nanostructured materials, energy-saving technologies, silicon-based structures, mathematical modelling, and other scientific areas, contributing to an environment in which research and education are closely interconnected.

KSU provides inclusive educational infrastructure to support access to university facilities for students and visitors with different needs. The university has accessible toilets, wheelchair ramps, tactile paving, Braille signboards, disabled parking spaces, and accessibility lifts. Such infrastructure supports equal participation in educational activities and improves accessibility across the university environment.

The university's campuses also provide facilities that support students' educational and personal development. The main campus includes 9 academic buildings, 8 dormitories, two libraries, indoor and outdoor sports facilities, and landscaped areas, while additional campuses provide further study blocks, dormitories, sports facilities, and student spaces. The availability of dormitory facilities and 96 computers in the Student Dormitory also supports students' access to educational resources outside formal classroom hours.

Student engagement is an important component of the university's educational approach. Between 2024 and 2025, student organizations conducted 10 major sustainability-related activities, while the number of student-led activities was 8 in 2023, 6 in 2024, and 4 in 2025, giving an average of approximately six activities per year. Through these initiatives, students develop leadership, teamwork, problem-solving, environmental awareness, communication, and practical project-management skills.

The university's educational activities also contribute to career development and employability. During the 2024/2025 academic year, 6,052 students graduated from the faculties covered in the report. Among them, 557 graduates obtained green jobs. Particularly notable results were recorded in Chemistry and Biology, with 130 graduates entering green jobs, Pedagogy with 120, and Geography and Agronomy with 100. These outcomes demonstrate the connection between academic programs, professional skills, and emerging employment opportunities in sustainability-related sectors.

Karshi State University utilizes digital learning management systems, including HEMIS and Moodle, together with other digital platforms, to support online, blended, and hybrid learning. These platforms facilitate access to course materials, learning resources, assignments, assessments, academic information, and communication between students and instructors. The integration of digital learning systems provides

students with greater flexibility in accessing educational content both on and off campus and supports the continuous digitalization of the teaching and learning process.

SDG 5 – Gender Equality

Achieve gender equality and empower all women and girls

Karshi State University has established a Council on Support of Women and Girls to coordinate initiatives promoting gender equality, preventing discrimination, and increasing women's participation in academic and social activities. The Council also cooperates with regional women's committees to identify talented female students and provide mentoring and guidance. In addition, the university has organised gender-audit meetings to discuss institutional responsibilities, review existing measures, and identify opportunities to strengthen equal rights and opportunities for women and men.

The university supports women's access to higher education through national policies and targeted assistance programmes. These include additional state grants for girls from low-income and vulnerable families, as well as full state-budget coverage of tuition fees for women enrolled in master's programmes at state universities. The university's support programme also identifies female students from disadvantaged backgrounds, including those with disabilities and those who have lost their breadwinners, for targeted financial assistance. These measures contribute to reducing financial barriers and promoting equitable access to educational opportunities.

Karshi State University promotes women's academic and social participation through mentoring and guidance coordinated by the Council on Support of Women and Girls. The university's programmes also include activities organised through the *Qizlarjon* club to encourage female students' social, cultural, and political engagement and support their personal development. Sports events and competitions for female students are also included in the university's planned activities, helping create opportunities for participation and engagement in university life.

The university has developed measures to identify and address harassment and violence affecting female students and staff. These include the "Only for Girls" Telegram channel for receiving concerns, regular surveys to identify harassment and violence, analysis of survey results, and individual responses based on identified needs. The university also plans explanatory sessions on national legislation protecting women from harassment and violence, including procedures for issuing protective orders. These measures demonstrate institutional attention to women's safety, rights, and access to support.

University programmes provide for free and anonymous psychological consultations for female students experiencing difficult circumstances. The Council on Support of Women and Girls, university psychologists, and relevant university officials are assigned responsibilities for identifying problems and coordinating assistance. Planned activities also include medical examinations and health-awareness initiatives in cooperation with regional healthcare organizations. Additional measures address the needs of women and girls from disadvantaged families and those requiring social, psychological, material, or rehabilitation support.

The university conducts awareness-raising activities to improve understanding of gender equality, women's rights, and equal opportunities. These include a roundtable discussion entitled "Ensuring Gender Equality: An Important Issue," a seminar-training on "Gender Equality Issues," and discussions involving students, academic staff, and women's advisers from different faculties. Participants are introduced to national legislation and reforms concerning women's rights, protection from violence, and participation in social, economic, and political life. Videos and presentations are also used to communicate information and encourage discussion of gender-related challenges.

Karshi State University has held meetings of gender auditors to discuss the university's responsibilities in promoting equality and implementing gender-related measures. The discussions have covered the roles of gender auditors, the responsibilities of the university as an employer, staff awareness, and the implementation of relevant policies and programmes. These activities provide a basis for reviewing institutional practices and identifying areas where additional action may be required to promote equal opportunities.

Karshi State University is committed to providing an inclusive, accessible, and supportive campus environment for students, academic staff, and visitors, including women with maternity-related needs. The university provides a dedicated maternity care room on campus, particularly at the Faculty of Primary Education, to support the comfort and well-being of women during maternity. In addition, the university operates a kindergarten on campus, helping support childcare needs and making it easier for university staff and other eligible members of the university community to balance their professional responsibilities with family life. These facilities demonstrate the university's practical commitment to creating a family-friendly environment and supporting women's participation in education and employment.

SDG 6 – Clean water and sanitation

Ensure availability and sustainable management of water and sanitation for all

Karshi State University operates 23 water reservoirs, including seven reservoirs with a capacity of 1,000 m³ each and 16 reservoirs with capacities ranging from 300 to 600 m³. The university obtains water from 15 artesian wells, providing a water supply for academic buildings, laboratories, administrative offices, student dormitories, and other campus facilities. The reservoir infrastructure supports water storage and distribution throughout the year.

In 2023, Karshi State University installed an ASMOS water purification and recycling station to improve water quality and promote sustainable water management. The system uses multi-stage filtration technology to purify groundwater from artesian wells for drinking and domestic purposes. With a processing capacity of 300 m³ per hour, the station supplies purified water for use across university facilities. Treated water is also used for non-potable purposes, including sanitation, cleaning, and irrigation, helping to reduce the use of drinking-quality water for routine campus operations.

The university implements water conservation measures to minimize waste and improve the efficient use of water resources. Green areas are irrigated using modern sprinkler systems and drip irrigation technologies, which help distribute water efficiently and reduce unnecessary consumption, particularly during dry periods. Non-potable technical water is used for irrigation to conserve drinking water. Rainwater collected through roof drainage systems is also reused to water green spaces, supporting sustainable landscape management.

Water-saving equipment is installed in university buildings, including low-flow taps and dual-flush toilets in restrooms and washrooms. These facilities help reduce unnecessary water consumption. Regular inspections, maintenance, and monitoring are carried out to ensure that water systems operate efficiently and to identify and prevent leaks.

Karshi State University implements measures to protect water quality and prevent contamination. The university’s water reservoirs are covered to reduce the risk of contamination from dust, debris, and external runoff. Four qualified specialists are responsible for monitoring and maintaining the ASMOS purification and water-storage system. Regular water-quality testing and technical inspections help prevent pollution and bacterial contamination and support the provision of safe water across campus.

All university buildings are connected to the regional main sewer system, with wastewater management coordinated in cooperation with Qashqadaryo Regional Suv Oqova. Approximately 60,000 m³ of wastewater is discharged annually from university facilities into the regional sewer network. Underground sewer pipelines are regularly maintained to prevent leakage, contamination, and blockages. This arrangement

supports the safe collection and management of wastewater in accordance with sanitary and environmental requirements.

The university's Environmental Sustainability Policy promotes responsible water management, efficient water use, and the prevention of water waste. Its priorities include implementing water conservation programs, introducing drip irrigation and wastewater recycling where feasible, and encouraging responsible water use through awareness campaigns and training. Students and staff are encouraged to adopt water-saving practices and contribute to the university's sustainability objectives.

During 2022–2024, Karshi State University allocated US$85,139.83 to water conservation, representing 1.4% of the sustainability budget allocation. These funds support initiatives such as efficient irrigation systems, water-saving fixtures, and awareness campaigns designed to reduce water consumption and strengthen sustainable resource management.

Karshi State University's water management initiatives contribute to SDG 6 through the provision of purified drinking water, water-quality monitoring, wastewater collection, efficient irrigation, rainwater reuse, water-saving equipment, and water conservation awareness. These measures support the responsible use of water resources, improved sanitation, pollution prevention, and the development of a more sustainable campus environment.

SDG 7 – Affordable and clean energy

Ensure access to affordable, reliable, sustainable and modern energy for all

Karshi State University is expanding its use of renewable energy through the installation of solar panels across its campus. A total of 2,790 solar panels have been installed on 17 university buildings, supported by a 510 kW on-campus solar power plant. The combined installed capacity of the university's solar energy systems is 1,545 kW. During the period from October 2024 to September 2025, the university generated 1,832,706 kWh of electricity from solar energy. Solar-generated electricity supplies campus facilities, reduces dependence on the regional electricity grid, and contributes to the university's transition towards cleaner energy. Surplus electricity is exported to the grid.

Solar energy plays a significant role in meeting the university's electricity needs. During the October 2024–September 2025 period, solar electricity generation represented a reported 54.3% of total electricity consumption, based on 1,832,706 kWh of solar generation and total electricity consumption of 3,375,993 kWh. The university's renewable energy infrastructure supports the efficient use of energy resources and helps

reduce greenhouse gas emissions associated with electricity supplied by the regional power grid.

Karshi State University implements energy-efficiency measures across its academic buildings, laboratories, administrative offices, and other facilities. All university buildings are equipped with LED lighting systems, and the replacement of traditional fluorescent and halogen lamps has contributed to electricity savings of up to 100,000 kWh annually. Thermal insulation has been installed in new academic and administrative buildings, helping reduce the energy required for heating and cooling. More than 90% of campus buildings are reported to have energy-saving technologies or insulation measures in place. The university also prioritizes energy-efficient electrical appliances and laboratory equipment. Energy efficiency is an important criterion in the procurement of computers, printers, servers, refrigerators, air-conditioning systems, and other equipment. Newly purchased appliances and laboratory instruments are selected from A+ and A++ energy-efficiency classes, as applicable.

The university uses an online energy-monitoring system to track electricity generation, consumption, and electricity exported to the grid. Energy use is monitored regularly to identify inefficiencies, improve resource management, and support evidence-based decisions on energy conservation. The campus also hosts a Laboratory for the Use of Alternative Energy Sources, where energy production, consumption, and savings are monitored using modern tracking technologies. These activities support the university’s efforts to improve energy performance and promote the practical application of renewable energy technologies.

The university’s electricity consumption has been reduced through solar energy generation, LED lighting, thermal insulation, energy-efficient equipment, and improved monitoring practices. Electricity is primarily used for lighting academic buildings and laboratories, operating heating and air-conditioning systems, powering computers and servers, and meeting the needs of student dormitories, cafeterias, and administrative offices. The university’s recorded electricity consumption decreased from 2,458,583 kWh in 2024 to 1,775,155 kWh in 2025, representing a reported reduction of 27.8%. This progress reflects ongoing efforts to improve energy efficiency and reduce dependence on purchased electricity.

Karshi State University’s Environmental Sustainability Policy promotes energy conservation, increased energy efficiency, the adoption of renewable energy sources where feasible, and the implementation of smart resource-management technologies. The university is committed to reducing energy consumption and greenhouse gas emissions through sustainable campus operations and responsible procurement. The Sustainability Committee oversees sustainability initiatives and supports the

implementation of the university's Green Campus objectives. Energy efficiency, renewable energy generation, and carbon management form important components of the university's long-term ambition to achieve carbon neutrality by 2030.

The university continues to use natural gas for heating, kitchens, student dormitories, and other campus needs. Recorded natural gas consumption for the October 2024–September 2025 period was 581,173 m³. Energy-related emissions are assessed using electricity consumption, solar energy generation, and natural gas use. The university's solar installations help offset emissions associated with electricity that would otherwise be supplied by the regional grid. Continued investment in renewable energy and energy-efficient technologies is intended to reduce the university's environmental footprint.

Karshi State University contributes to SDG 7 – Affordable and Clean Energy through renewable electricity generation, solar infrastructure development, energy-efficient lighting, building insulation, efficient equipment procurement, and digital energy monitoring. These initiatives support more responsible energy consumption, reduce reliance on grid electricity, and strengthen the university's commitment to a cleaner, more energy-efficient, and sustainable campus.

SDG 8 – Decent work and economic growth

Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

Karshi State University supports human capital development through higher education, professional training, research, innovation and employee welfare. The University promotes graduate employability, encourages student entrepreneurship, supports scientific and technological development, and invests in sustainable infrastructure and institutional operations.

The University employs 812 lecturers and researchers across multiple academic disciplines. Through its educational programs, research activities and cooperation with employers, industry representatives, academic experts and international partners, KSU works to strengthen the relevance of curricula and improve graduates' readiness for the labour market. During the 2024/2025 academic year, the University recorded 6,052 graduates, including 557 graduates reported as having green jobs. These figures highlight the role of higher education in developing skills relevant to emerging employment opportunities and the green economy.

KSU implements social support and incentive programs for employees and socially vulnerable groups. These initiatives contribute to staff welfare and institutional support

for members of the university community. In 2025, the University allocated more than USD 2.4 million for social support and incentive programs.

The University supports scientific and technological development through research activities, innovation initiatives and student engagement in sustainability-oriented projects. The *Eko-Innovation 2025* competition encouraged students to develop ideas related to environmental protection, waste recycling, climate change mitigation, renewable energy, energy efficiency and sustainable materials. The competition offered financial awards of UZS 6 million, UZS 4 million and UZS 2 million for the first, second and third places, respectively. KSU also promotes research in areas such as the green economy, sustainable agriculture, renewable energy and resource management. These activities support the development of knowledge and practical skills relevant to innovation and sustainable economic growth.

The University invests in sustainable infrastructure, energy efficiency and responsible resource management to improve institutional operations. Initiatives include renewable energy generation, energy-saving technologies, waste reduction and recycling, and the reuse of organic waste. These measures support more efficient use of resources and contribute to the development of environmentally responsible institutional practices.

Indicator	Result
Lecturers and researchers	812
Graduates in the 2024/2025 academic year	6,052
Graduates reported as having green jobs	557
Social support and incentive allocation in 2025	More than USD 2.4 million
Google Scholar publications in 2024	745
Scopus publications in 2024	267

Karshi State University contributes to SDG 8 by developing human capital, supporting employee welfare, strengthening graduate employability, promoting student innovation and advancing research. Its investments in sustainable infrastructure and resource efficiency complement these efforts by supporting responsible institutional development.

SDG 9 – Industry, innovation and infrastructure

Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation

Karshi State University promotes scientific research, technological development, digital transformation and sustainable infrastructure. Through investments in ICT

systems, renewable energy, research activities and student innovation, the University supports the development of modern educational and research environments.

KSU supports interdisciplinary research and scientific collaboration in areas relevant to technological development and sustainability. Research priorities include renewable energy, environmental protection, sustainable agriculture, water-resource management, climate change mitigation and the green economy. The University encourages academic staff to publish in scientific journals, participate in international research networks and apply for research grants. Student-led innovation is also supported through initiatives that encourage practical solutions to environmental and technological challenges.

Indicator	Result
Google Scholar publications in 2024	745
Scopus publications in 2024	267
Lecturers and researchers	812

KSU promotes student innovation through the Eko-Innovation 2025 competition, which encourages the development of practical solutions to environmental challenges.

The competition focuses on three areas:

- Environmental protection, waste recycling and climate change mitigation.
- Renewable energy and energy-saving technologies.
- Innovative ecological devices and sustainable materials.

Undergraduate, master's and doctoral students aged 18–35 can participate individually or in teams. The competition provides financial awards of UZS 6,000,000 for first place, UZS 4,000,000 for second place and UZS 2,000,000 for third place.

The University has developed digital infrastructure to support teaching, research, communication and administrative management. Classrooms and academic facilities use computers, projectors, interactive panels, smart televisions and other multimedia technologies. Digital platforms, including Zoom, facilitate online teaching, academic meetings and international collaboration.

The University's ICT infrastructure includes:

Indicator	**Quantity**
Computers	1,305
Video projectors	128
Interactive panels	44
Smart televisions	103
Wi-Fi routers	152
Information kiosks	10

LED screens measuring 6 × 4 m	2
LED screens measuring 3 × 2 m	7
E-MINBAR systems	4
Classrooms equipped with IP surveillance cameras	157
Computer laboratories	16
Computers in the Information Resource Centre	140
Computers in the student dormitory	96

The University's ICT systems are interconnected through an internal digital management network that supports administrative planning, infrastructure monitoring and performance evaluation.

KSU invests in renewable energy and energy-efficient technologies to modernize campus infrastructure and improve resource management.

Indicator	Result
Solar panels installed	2,790
Buildings with solar panels	17
Capacity of the campus solar power station	510 kW
Combined capacity of solar systems	1,545 kW
Solar electricity generated, October 2024–September 2025	1,832,706 kWh
Total electricity consumption over the same 12-month period	3,375,993 kWh
Reported share of electricity consumption covered by solar generation	54.3%

Karshi State University operates three campus sites with a combined area of 319,107 m². The campuses provide facilities for education, research, student accommodation and other university activities.

Infrastructure indicator	Result
Total campus area	319,107 m²
Main Campus area	285,356.83 m²
Philology Faculty Campus area	24,278.17 m²
3rd Micro-District Campus area	9,472 m²
Academic buildings on the Main Campus	9
Dormitories on the Main Campus	8
Libraries on the Main Campus	2
Computer laboratories	16

The campuses also include study blocks, indoor and outdoor sports facilities, libraries and landscaped areas that support academic and student activities.

KSU promotes the efficient use of resources through infrastructure modernization, digital management and environmentally responsible practices.

Key initiatives include:

- Energy management: online monitoring of electricity consumption and renewable-energy generation.
- Water conservation: promotion of efficient water use, drip irrigation and wastewater recycling where feasible.
- Waste management: waste segregation, recycling and cooperation with waste-management organizations.
- Digitalization: use of digital platforms and administrative systems to improve operational efficiency and reduce paper consumption.
- Sustainable procurement: prioritization of energy-efficient equipment and environmentally responsible products.

These initiatives support the modernization of university operations and the development of more resource-efficient infrastructure.

Karshi State University contributes to SDG 9 through scientific research, student innovation, digital transformation and investment in modern infrastructure. Its ICT facilities support education and research, while renewable-energy installations and energy-monitoring systems demonstrate the application of technology to institutional sustainability. Through research development, innovation-focused activities and infrastructure modernization, KSU seeks to strengthen its contribution to technological progress and sustainable development.

Documented indicators and results

- Sustainability research funding, 2025: 2,390,005,500 UZS
- Sustainability research funding, 2024: 2,439,188,737 UZS
- Sustainability research funding, 2023: 1,403,660,563 UZS
- Sustainability-related courses: 169

SDG 10 – Reduced Inequalities

Reduce inequality within and among countries

Karshi State University is committed to promoting an inclusive and accessible educational environment that supports equal access to university facilities for students, employees, and visitors. The university implements infrastructure measures to improve accessibility for people with disabilities and individuals with special needs, supporting mobility, independence, comfort, and participation in campus life.

The university’s Code of Ethics establishes principles of ethical conduct, mutual respect, accountability, and responsible behavior within the academic community.

Karshi State University promotes an environment in which students and employees are treated fairly and can participate in academic, professional, and social activities with dignity.

The university's equality and inclusion commitments address discrimination based on personal characteristics, including sex, age, disability, nationality, ethnic origin, religion or belief, pregnancy and maternity, and other relevant circumstances.

Further information is available in the Code of Ethics.

Karshi State University provides a range of facilities to improve campus accessibility, including:

- Designated parking spaces for people with disabilities.
- Accessible toilets in major university buildings, including faculties, student dormitories, the rectorate, and the library.
- Wheelchair ramps at main building entrances and along campus pathways.
- Tactile paving on corridors and key pedestrian routes to support people with visual impairments.
- Braille signage on doors and directional signs to facilitate navigation.
- Wheelchair lifts and platform lifts in selected multi-storey buildings to improve access between floors.

These facilities contribute to a more inclusive campus environment and help reduce physical barriers to university services and infrastructure.

Karshi State University maintains safety and security arrangements across its academic buildings, student dormitories, and outdoor areas. These measures include CCTV surveillance, an emergency hotline and alarm system, and trained security personnel. Such provisions contribute to a safer campus environment for students, academic staff, administrative employees, and visitors, including individuals who may require additional support.

The university incorporates accessibility considerations into the maintenance and improvement of its facilities. Accessible entrances, pedestrian routes, sanitary facilities, signage, and vertical mobility equipment support safer and more convenient movement throughout the campus.

Accessibility measure	Provision at Karshi State University
Designated disability parking	Available
Accessible toilets	Available in major university buildings
Wheelchair ramps	Installed at main entrances and along pathways
Tactile paving	Available on corridors and key pedestrian routes
Braille signage	Installed on doors and directional signs

Accessibility lifts	Available in selected multi-storey buildings

The university's institutional commitments include fair working conditions, occupational safety, social protection, and respect for fundamental labour rights. The university also recognizes the importance of equal pay for work of equal value and monitoring remuneration to identify and address potential gender-based pay disparities.

Employees are entitled to raise concerns regarding labour rights, remuneration, working conditions, and other employment-related matters through formal complaint procedures. Transparent and confidential grievance mechanisms help strengthen accountability and support a fair working environment.

Karshi State University recognizes the importance of protecting the rights and wellbeing of employees during pregnancy and maternity. Its institutional commitments include maternity and childcare leave, job protection, and flexible or adapted working conditions where necessary. These measures contribute to a more supportive working environment and help promote equal opportunities for employees with family responsibilities.

The university promotes inclusion through educational activities that connect academic learning with the needs of diverse communities:

In November 2024, the Faculty of Pedagogy organized a visit to Specialized Boarding School No. 17 in Karshi for children who are deaf or hard of hearing. Psychology students, accompanied by the Dean's Advisor on Women's Issues, participated in activities designed to develop practical experience, communication skills, empathy, and understanding of inclusive education.

The initiative connected university learning with community engagement and helped students develop awareness of the needs and experiences of children with special educational needs:

In February 2025, representatives of the university's Incubation Center visited secondary schools No. 1 and No. 7 in Nishon district to promote STEAM education, digital literacy, programming skills, and career awareness.

The outreach activities also encouraged girls' participation in information technology and vocational fields, helping broaden access to educational opportunities and future career pathways.

SDG 11 – Sustainable cities and communities

Make cities and human settlements inclusive, safe, resilient and sustainable

Karshi State University is committed to promoting sustainable, inclusive, safe, and environmentally responsible communities through green campus development,

sustainable mobility, waste reduction, environmental education, and community engagement. Located in Karshi, Kashkadarya Region, the university integrates sustainability into its infrastructure, daily operations, research, and partnerships with local organizations. These initiatives contribute to improving the campus environment, encouraging responsible resource use, reducing environmental impacts, and promoting a culture of sustainability within the university and the surrounding community.

Karshi State University operates three campus sites within the urban area of Karshi, integrating academic facilities, student accommodation, libraries, sports facilities, and landscaped green spaces.

Indicator	Value
Total campus area	319,107 m²
Main campus area	285,356.83 m²
Philology Faculty campus area	24,278.17 m²
3rd Micro-District campus area	9,472 m²
Total building area across the three campuses	116,897.76 m²
Total planted vegetation area	177,328 m²
Share of campus area covered by planted vegetation	Appr. 55.6%
Forest vegetation area used for research, teaching, and community engagement	44,332 m²
Permeable surfaces supporting natural drainage	38,472 m²
Share of campus area represented by permeable surfaces	12%

The university's green spaces support biodiversity conservation, environmental education, temperature regulation, and carbon absorption. Permeable surfaces help rainwater enter the ground and improve natural drainage, while planted vegetation helps reduce heat accumulation. The campuses also provide spaces for recreation, physical activity, and academic engagement with the natural environment. The 3rd Micro-District campus includes a small park, contributing to the presence of green areas within the university's urban setting.

Karshi State University promotes sustainable campus mobility by prioritizing walking, cycling, and electric transport. The university has introduced restrictions on private motor vehicles to reduce traffic within the campus and create a safer environment for pedestrians and cyclists.

Key measures include:

- A 544-metre pedestrian and bicycle path connecting the university's two main gates.
- Approximately 8,000 metres of sidewalks on the main campus.

- Dedicated pedestrian routes and bicycle paths connecting academic buildings, dormitories, laboratories, libraries, and other facilities.
- Restrictions on private diesel and gasoline vehicles within the campus, with exceptions for authorized emergency and service access.
- Closure of an internal parking area of approximately 1,200 m².
- Two electric golf shuttles, each with a capacity of 10 passengers.
- Seven electric motorcycles used for technical and maintenance operations.
- A 36-seat university bus used for student academic, cultural, and other university-related trips.

The closure of the internal parking area addresses vehicle movements and idling that were previously estimated to generate approximately 1.2 tonnes of carbon dioxide annually. The initiative supports lower-emission mobility and helps reduce vehicle-related noise and air pollution within the campus. The university has also considered an app-based sharing system involving 20 electric scooters as a potential extension of its sustainable mobility arrangements.

Karshi State University implements a waste-management programme based on the principles of Reduce, Reuse, and Recycle (3R). Separate waste containers for paper, plastic, and organic waste are installed near faculty buildings, supporting waste separation at the source.

The university collaborates with Eco Trans for the collection and transfer of recyclable materials and promotes responsible waste management through environmental campaigns and student participation.

Paper and plastic waste reduction

Waste category	Previous year (tonnes)	2025 (tonnes)	Reduction (tonnes)
Plastic	45.3	29.2	16.1
Paper	73.3	47.4	25.9
Total	118.6	76.6	42.0

The university reduces paper consumption through digital administrative and educational processes, electronic communication, and double-sided printing. Its cafeterias have replaced single-use plastic containers with reusable ceramic or metal tableware. Student dormitories provide filtered drinking water as an alternative to plastic water bottles.

Organic waste management

Indicator	Previous year (tonnes)	2025 (tonnes)	Reduction (tonnes)
Total organic waste	352.7	227.6	125.1

Food waste	317.4	209.3	108.1
Leaf waste	35.3	18.3	17.0

Organic waste is collected separately and processed for reuse where possible.

- 18.7 tonnes of food waste were processed and reused as animal feed.
- 18 tonnes of leaf waste were converted into nutrient-rich biomass fertilizer using California red worms.

These practices support more responsible resource use and reduce the amount of organic waste requiring disposal.

The student volunteer group Yosh Ekopatrul participates in clean-up campaigns both on campus and throughout Karshi city. The group also supports environmental awareness activities and responsible waste-sorting practices. The university organizes Eco Week activities in cooperation with the Kashkadarya Regional Ecology Department, including environmental events, training sessions, public-awareness campaigns, and waste collection and sorting activities. These initiatives extend environmental responsibility beyond the university campus and encourage students to contribute to cleaner urban surroundings.

The university invests in energy-efficient buildings, renewable energy, and green infrastructure to reduce environmental impacts and improve the sustainability of campus operations.

Indicator	Value
University buildings equipped with LED lighting	100%
Buildings reported as insulated	90%
Solar panels installed	2,790
Buildings with solar panels	17
Small campus solar power plant capacity	510 kW
Total capacity of solar energy systems	1,545 kW
Solar electricity generated in 2024–2025	1,832,706 kWh
Campus area covered by greenery	177,328 m²
Permeable surfaces supporting natural drainage	38,472 m²

Energy-efficient lighting and building insulation help reduce electricity consumption for lighting, heating, and cooling. Renewable energy systems reduce dependence on electricity purchased from the regional grid. Green spaces and permeable surfaces also support local environmental quality by helping regulate temperature, reduce heat accumulation, and improve natural drainage. Karshi State University has established a long-term strategy to achieve carbon neutrality by 2030.

The university maintains safety and security arrangements intended to protect students, employees, and visitors across academic buildings, dormitories, and outdoor areas.

The campus safety infrastructure includes:

- CCTV surveillance systems.
- An emergency hotline and alarm system.
- Trained security personnel.
- Fire extinguishers, hydrants, and fire alarm systems.
- Regular inspection and maintenance of fire-safety equipment.
- Safety drills and awareness programs.
- A reported emergency response time of less than five minutes for accidents, crimes, fires, and other emergencies.

These measures support a safer learning and working environment and strengthen preparedness for emergencies within the university community.

Karshi State University promotes sustainability through educational activities, research, student volunteering, and cooperation with local organizations.

Key initiatives include:

- Eco Week: environmental awareness events, waste-separation training, and campaigns organized in cooperation with the Kashkadarya Regional Ecology Department.
- Environmental education conference: an international scientific and practical conference on “Environmental Education and Sustainable Development,” attended by more than 100 scientists and students.
- Yosh Eco-patrul: student-led clean-up activities on campus and in Karshi city.
- Sustainability Committee: established on 3 April 2025 to coordinate Green Campus activities and promote environmental awareness.
- Greenhouse and agricultural research: facilities for cultivating tomatoes, cucumbers, edible and medicinal mushrooms, and potatoes, supporting practical education in sustainable horticulture and agriculture.

Through these activities, the university encourages students and staff to take an active role in environmental protection and strengthens cooperation with local stakeholders.

The university’s Environmental Sustainability Policy promotes responsible resource management, waste reduction, sustainable transportation, environmental education, and cooperation with local communities and government agencies. The Sustainability

Committee coordinates sustainability-related activities and encourages participation by faculties, students, and administrative units.

For 2022–2024, the university recorded the following financial allocations:

Indicator	Amount
Total institutional budget	USD 61,597,422
Sustainability-related budget	USD 18,479,226
Reported sustainability share of total budget	30%
Sustainable transport allocation	USD 34,056.37
Facility maintenance for sustainability	USD 4,894,681.86

Investment in facility maintenance supports building upgrades, energy efficiency, accessibility, and green infrastructure standards. These measures contribute to the long-term sustainability of university facilities and their surrounding environment.

SDG 12 – Responsible consumption and production

Ensure sustainable consumption and production patterns

Karshi State University promotes responsible consumption and production through waste prevention, recycling, reuse of materials, sustainable procurement, digitalization, water management, and environmental education. The university applies the principles of Reduce, Reuse, and Recycle (3R) across campus operations and works with external waste-management partners to improve resource efficiency and reduce environmental impacts.

The university has established a waste-separation system with designated containers for paper, plastic, and organic waste near faculty buildings. Collected recyclable materials are transferred to a central collection area, weighed, documented, and sent for recycling in cooperation with Eco-Trans, a licensed waste collection and disposal organization in Uzbekistan.

Paper and plastic waste

Waste category	Previous year (tonnes)	2025 (tonnes)	Reduction (tonnes)
Paper	73.3	47.4	25.9
Plastic	45.3	29.2	16.1
Total	118.6	76.6	42.0

In 2025, paper and plastic waste totalled 76.6 tonnes, a reduction of 42 tonnes compared with the previous year. These efforts support more efficient material use and improved waste management across academic and administrative buildings.

The university separates organic waste generated by cafeterias, student dormitories, and campus green areas. Food and leaf waste are collected and processed for reuse where possible.

Organic waste category	Previous year (tonnes)	2025 (tonnes)	Reduction (tonnes)
Food waste	317.4	209.3	108.1
Leaf waste	35.3	18.3	17.0
Total organic waste	352.7	227.6	125.1

The university's organic waste management initiatives include:

- 18.7 tonnes of food waste processed and reused as animal feed through granulation.
- 18 tonnes of leaf waste converted into nutrient-rich biomass fertilizer using California red worms.
- Separate collection, storage, and processing of organic materials.
- Cooperation with Eco Trans for the appropriate management of remaining waste.

These practices reduce waste requiring disposal and promote circular resource use by returning useful materials to productive applications.

The university monitors inorganic waste generation and promotes the reuse and recovery of construction materials.

Waste category	Previous year (tonnes)	2025 (tonnes)	Reduction (tonnes)	Reported reuse or recovery
Total non-toxic inorganic waste	381.0	245.7	135.3	—
Construction materials, including metals	262.4	169.1	93.3	52.7 tonnes upcycled
Paper	73.3	47.4	25.9	Sent for recycling
Plastic	45.3	29.2	16.1	Sent for recycling

The figures show a reduction in non-toxic inorganic waste from 381 tonnes to 245.7 tonnes. Construction waste management includes material recovery and the reuse of suitable materials, supporting more efficient use of resources and reduced disposal requirements.

Karshi State University applies specific handling procedures to toxic waste generated by laboratory activities, medical training units, and maintenance facilities.

Indicator	Previous year	2025	Reduction
Toxic waste generated	1.7 tonnes	1.1 tonnes	0.6 tonnes

Hazardous waste management measures include:

- Separation and appropriate labelling of hazardous materials.
- Temporary storage under controlled conditions.
- Safe internal collection and transportation.
- External collection and treatment through Eco Trans.
- Regular inspections and staff training.

The Department of Maintenance and Environmental Management oversees these processes to support safe handling, reduce hazardous waste generation, and prevent environmental contamination.

The university promotes responsible consumption by reducing disposable materials and encouraging reusable alternatives.

Key initiatives include:

- Replacing single-use plastic containers in university cafeterias with reusable ceramic or metal tableware.
- Providing filtered drinking water in student dormitories as an alternative to plastic water bottles.
- Encouraging double-sided printing and reducing unnecessary paper use.
- Promoting electronic communication and digital administrative processes.
- Raising awareness of responsible waste disposal and material reuse.

These measures encourage students and employees to adopt more sustainable consumption habits in their daily activities.

Karshi State University uses digital platforms to reduce paper consumption and improve administrative efficiency.

- The HEMIS system is used to manage class schedules, orders, reports, and student gradebooks electronically.
- Google Classroom and Moodle support digital teaching and learning.
- Electronic communication reduces the need for printed documents.
- Internal digital records track waste generation, collection, and recycling.
- The ICT Department and the Department of Maintenance and Environmental Management analyze waste data to evaluate performance and plan reduction measures.

Digitalization supports more efficient resource use while helping the university monitor its waste-management activities.

The university's Environmental Sustainability Policy prioritizes environmentally friendly and ethically sourced products, sustainable food and catering services, responsible printing, and efficient resource management.

For 2022–2024, the university reported the following financial allocations related to sustainability:

Indicator	Amount
Total institutional budget	USD 61,597,422
Sustainability-related budget	USD 18,479,226
Sustainable procurement allocation	USD 996,338.46

Sustainable procurement supports the selection of environmentally friendly and locally sourced materials for campus operations and maintenance. The university also identifies energy-efficient equipment, reduced paper consumption, and sustainable catering practices as priorities.

The reported sustainable procurement allocation is part of the university's sustainability-related expenditure and should not be interpreted as expenditure exclusively on waste reduction.

Responsible water use and wastewater management form part of the university's resource-efficiency commitments.

- All university buildings are connected to the regional sewer network.
- Approximately 60,000 m^3 of wastewater are discharged annually into the regional sewer network.
- Cooperation with the Kashqadaryo Regional Water and Wastewater organization supports wastewater collection and treatment.
- Underground sewer pipelines are maintained to help prevent leakage, contamination, and blockages.
- The university's environmental policy promotes water conservation, efficient irrigation, and wastewater recycling where feasible.

These measures support responsible management of water-related waste streams. The stated wastewater volume represents discharge into the regional sewer network, not a quantified volume of wastewater recycled by the university.

Karshi State University promotes responsible consumption through awareness campaigns, student participation, and cooperation with local environmental organizations.

Activities include:

- Eco Week, featuring waste-separation training, environmental events, and awareness campaigns.

- Cooperation with the Kashkadarya Regional Ecology Department on environmental education.
- Student-led clean-up activities organized by the volunteer group Yosh Ekopatrul.
- Campaigns promoting reduced paper and plastic consumption.
- Seminars such as “Save Nature,” “Ecological Culture – Sustainable Future,” and “Green Uzbekistan.”
- An international scientific and practical conference on “Environmental Education and Sustainable Development,” attended by more than 100 scientists and students.

These initiatives encourage responsible consumption habits and strengthen awareness of waste prevention, recycling, and environmental protection.

The university uses digital records and data analysis to monitor waste generation, track recycling activities, and evaluate progress under the 3R approach. Waste collection and recycling data are maintained through internal digital platforms. The ICT Department and the Department of Maintenance and Environmental Management jointly analyze information and prepare annual assessments to identify opportunities for improvement. Cooperation with Eco Trans also supports the monitoring of waste collection and disposal processes.

SDG 13 – Climate action

Take urgent action to combat climate change and its impacts

Karshi State University is committed to addressing climate change through renewable energy development, energy efficiency, greenhouse gas emissions reduction, sustainable transport, and environmental management. The university promotes practical measures to reduce its carbon footprint, improve energy performance, and strengthen climate responsibility across its campuses.

Karshi State University monitors energy consumption and estimates greenhouse gas emissions associated with electricity and natural gas use. During October 2024–September 2025, the university recorded the following energy-related indicators:

Indicator	Value
Electricity purchased from the grid	1,543,287 kWh
Electricity generated by solar panels	1,832,706 kWh
Natural gas consumption	581,173 m^3
Electricity emission factor	0.4 kg CO_2/kWh
Natural gas emission factor	1.9 kg CO_2/m^3
Reported emissions from grid electricity	617.31 tonnes CO_2
Reported emissions avoided through solar generation	733.08 tonnes CO_2

Reported emissions from natural gas	1,141.9 tonnes CO_2
Reported net carbon footprint	1,759.21 tonnes CO_2e

The university aims to reduce emissions through renewable energy generation, energy-efficient infrastructure, improved energy management, and sustainable transport practices.

Karshi State University is expanding its use of renewable energy to reduce dependence on conventional electricity sources and support low-carbon campus operations.

Key indicators include:

- 2,790 solar panels installed across university facilities.
- Solar panels installed on 17 university buildings.
- A campus solar power station with a capacity of 510 kW.
- Total installed solar capacity of 1,545 kW.
- Solar electricity generation of 1,832,706 kWh during October 2024–September 2025.
- Total electricity consumption of 3,375,993 kWh over the corresponding reported 12-month period.
- A reported solar contribution of 54.3% of total electricity consumption.

Solar generation helps reduce reliance on grid electricity and contributes to the university's long-term emissions-reduction objectives.

The university implements energy-saving measures to improve building performance and reduce unnecessary energy consumption.

Measure	Implementation and results
LED lighting	LED lamps installed throughout 100% of university buildings
Annual electricity savings	Up to 100,000 kWh through LED lighting
Thermal insulation	Approximately 90% of buildings reported as thermally insulated
Energy savings from insulation	Reported reduction in energy consumption of 20–25%
Energy-saving technologies	Installed in more than 90% of buildings
Energy-efficient computers	1,000 out of 1,305 computers (77%)
Energy-efficient air conditioners	268 out of 383 units (70%)
Energy-efficient refrigerators and freezers	45 out of 65 units (69%)
Energy-efficient laboratory equipment	160 out of 220 units (73%)
Energy-efficient lighting	11,445 out of 11,445 lamps (100%)

equipment	

The university also prioritises purchasing equipment with A+ and A++ energy-efficiency ratings. Energy consumption is monitored to identify opportunities for further savings and improve the efficiency of campus operations.

Karshi State University promotes low-emission mobility and safer pedestrian movement across campus.

Initiatives include:

- A 544-metre pedestrian and bicycle route connecting two main campus gates.
- Restrictions on private motor vehicle access to campus, with exceptions for emergencies.
- Closure of an internal parking area of approximately 1,200 m² as part of efforts to reduce unnecessary vehicle movement and idling.
- Use of electric transport for selected university operations, including electric motorcycles and electric golf shuttles.
- Promotion of walking and cycling as practical alternatives to short vehicle journeys.

The university has estimated that vehicle movements and idling at the former parking area contributed approximately 1.2 tonnes of CO_2 annually. Closing the area is intended to help reduce these emissions.

The university maintains green areas and permeable surfaces that support environmental quality and climate resilience.

Indicator	Area	Share of campus
Vegetated areas	177,328 m²	55.6%
Permeable surfaces	38,472 m²	12%

Green spaces support carbon absorption, help regulate local temperatures, reduce heat accumulation, and contribute to rainwater infiltration. Permeable surfaces also help reduce surface runoff and support more sustainable campus land management.

The university strengthens institutional responsibility for sustainability and climate action through dedicated governance arrangements. The Sustainability Committee, established by a University Council decision on 3 April 2025, coordinates sustainability-related activities and supports the integration of environmental considerations into university operations. The university's long-term objective is to achieve carbon neutrality by 2030. Progress towards this goal is supported by renewable energy development, energy conservation, emissions monitoring, sustainable transport, green campus management, and environmental awareness initiatives.

Karshi State University promotes environmental responsibility through sustainability activities, energy monitoring, and educational initiatives. Solar energy facilities support the monitoring of electricity generation, consumption, and energy savings. Digital monitoring of energy use helps identify opportunities to improve operational efficiency. The university also encourages students and staff to participate in environmental activities and develop awareness of climate change, resource conservation, and sustainable development.

SDG 14 – Life Below Water

Conserve and sustainably use the oceans, seas and marine resources

Although Karshi State University (KSU) is a land-based university and does not operate in a coastal or marine environment, its environmental sustainability activities contribute indirectly to the prevention of water pollution and the responsible use of freshwater resources. These activities are relevant to SDG 14 through pollution prevention, wastewater management, water-quality monitoring and environmental awareness.

KSU's water-management practices include closed water-management systems and wastewater treatment measures**,** which help reduce the risk of uncontrolled discharge and associated pollution of water systems. Water quality is monitored through the University's water purification and storage systems, supporting responsible water use and the maintenance of appropriate water quality for campus needs.

The University also conducts environmental education and awareness activities addressing pollution prevention, responsible water use, waste management and broader environmental sustainability. These activities contribute to building awareness of the relationship between land-based pollution and the protection of aquatic ecosystems.

The university operates a closed water management system designed to protect water resources from contamination.

Key measures include:

- 23 covered water reservoirs that help prevent contamination from dust, debris, and surface runoff.
- 15 artesian wells supplying groundwater for university needs.
- Water purification using ASMOS technology to produce safe drinking water.
- Regular water-quality testing and monitoring by four qualified specialists.
- Routine inspections and technical maintenance to reduce the risk of contamination and accidental pollution.
- Preventive maintenance of water infrastructure to reduce leaks and uncontrolled releases.

These measures help protect groundwater quality and minimize pollution risks associated with campus operations.

All university buildings are connected to the regional municipal sewer network. Wastewater generated on campus is transferred to the regional system for treatment, preventing direct discharge into the surrounding environment.

Indicator	Information
Annual wastewater discharge into the regional sewer network	Approximately 60,000 m³
Regional wastewater management partner	Qashqadaryo viloyat suv oqova
Sewer infrastructure	Underground pipelines connecting university buildings to the regional network
Preventive measures	Regular pipeline maintenance, inspections, and technical monitoring
Main objective	Prevent leakage, contamination, and uncontrolled wastewater discharge

Cooperation with the regional wastewater organization supports responsible sewage management and compliance with applicable sanitary and environmental requirements.

Karshi State University implements measures to conserve water and reduce the risk of environmental contamination.

- Non-potable technical water is used to irrigate green areas, reducing the consumption of drinking water.
- Rainwater collected through roof drainage systems is reused for watering green spaces.
- Water reservoirs and distribution infrastructure are regularly maintained.
- Water quality is monitored to prevent contamination and bacterial pollution.
- Students and staff are encouraged to adopt responsible water-use practices.

These activities support the protection of water resources and help minimize the potential environmental impacts of campus operations.

SDG 15 – Life on land

Protect, restore and promote sustainable use of terrestrial ecosystems

Karshi State University promotes the conservation of terrestrial ecosystems, biodiversity, forests, and green spaces through tree-planting campaigns, land conservation initiatives, environmental education, and sustainable campus management. The university works with environmental authorities, academic departments, and

student volunteers to protect local habitats and encourage the responsible use of land and natural resources.

The university maintains green areas and forest zones that support biodiversity, environmental education, and research.

Indicator	Value
Planted vegetation and green areas	177,328 m²
Forest area designated for research, teaching, and community engagement	Appr. 44,000 m²
Green-space campaigns conducted in 2024	5
Trees planted during the 2024 campus campaigns	More than 500
Flower species planted during the 2024 campaigns	30 species
Trees planted through the “Ziyolilar Bog’i” campaign in Muborak district	More than 1,000

The university’s Green Campus initiative supports the development and maintenance of green spaces. These areas contribute to local environmental quality and provide opportunities for ecological research, practical learning, and biodiversity awareness.

Karshi State University supports the conservation of local ecosystems through tree planting, forest conservation, landscaping, and environmental protection activities.

Key initiatives include:

- Maintaining planted green areas that provide habitats for local plant and animal species.
- Conducting tree-planting and greening campaigns to expand vegetation and restore degraded land.
- Organizing activities to clean and protect natural habitats in and around the university campus.
- Involving students and academic departments in biodiversity assessments and environmental awareness initiatives.
- Cooperating with the Department of Ecology, Environmental Protection and Climate Change on conservation activities.

The university also works to integrate biodiversity considerations into campus development and infrastructure planning.

The university has established procedures aimed at identifying, monitoring, and protecting species listed in the IUCN Red List and the National Red Book of Uzbekistan within and around its campus. These activities involve cooperation with environmental authorities and participation by students and academic staff from the Biology and Geography departments. Monitoring of forest and green areas, biodiversity assessments, awareness campaigns, and habitat maintenance support these conservation efforts.

Campus planning and construction activities are also intended to avoid adverse effects on existing trees, green spaces, and local habitats.

Karshi State University promotes biodiversity-friendly landscaping and the responsible selection of plant species.

Measures include:

- Prioritizing native and climate-adapted plant species in tree planting and landscaping projects.
- Selecting drought-resistant plants for post-construction landscaping where appropriate.
- Monitoring campus vegetation and removing invasive plants where identified.
- Educating students and staff about the environmental impacts of alien and invasive species.
- Involving students in conservation, restoration, and environmental monitoring activities.

These measures help maintain ecological balance and support the protection of local biodiversity.

Karshi State University integrates ecosystem conservation and sustainable land management into teaching, research, and community engagement. The university provides 169 courses across multiple faculties, including Geography, Chemistry, Engineering, and Biology, incorporating topics such as applied ecology, geoecology, environmental conservation, biogeography, biotechnology, and sustainable natural resource use.

Environmental education and outreach activities include:

- Open lectures, seminars, and practical workshops on environmental protection and wildlife conservation.
- Environmental awareness events for World Wildlife Day, Bird Day, and Earth Day.
- The “Eco-Tourism Zakovat” intellectual quiz for students of Ecology and Environmental Protection.
- The “Ecology Experts” quiz.
- The “Young Ecologists” Movement and regional environmental meetings.
- Tree-planting and habitat-cleaning activities involving students and local communities.

Between 2023 and 2025, the university conducted or participated in more than 15 environmental improvement campaigns across Karshi and the Kashkadarya region.

During 2024–2025, it also organized and supported more than 30 sustainability-related events.

The student volunteer group Yosh Ekopatrul plays an active role in the university's environmental initiatives.

Activities include:

- Participating in tree-planting and green-space campaigns.
- Supporting habitat maintenance and conservation activities.
- Organizing and participating in environmental clean-ups.
- Raising awareness of biodiversity protection and sustainable lifestyles.
- Contributing to reforestation and campus greening efforts.

The university also encourages students to develop innovative environmental solutions through initiatives such as the Eko-Innovation – 2025 competition, which supports ideas related to environmental protection, recycling, climate change mitigation, and sustainable technologies.

The university integrates biodiversity conservation into campus planning and development processes. Before new buildings or infrastructure projects begin, the Department of Maintenance and Environmental Management works with the Faculty of Geography and Earth Sciences and the Department of Ecology to assess potential environmental impacts. The approach includes preserving existing trees and green spaces where possible and prioritizing native and drought-resistant plants in landscaping after construction. These measures aim to protect local biodiversity while supporting the university's infrastructure development.

SDG 16 – Peace, Justice and Strong Institutions

Promote peaceful and inclusive societies, provide access to justice for all and build effective, accountable and inclusive institutions

Karshi State University (KSU) has established institutional mechanisms supporting integrity, ethical conduct, anti-corruption, stakeholder participation, complaints management, accountability and information security**.** These mechanisms are designed to promote a transparent, inclusive and responsible institutional environment for students, employees, researchers and other stakeholders.

The university coordinates its sustainability activities through a cross-functional Sustainability Committee that brings together representatives of academic faculties, student organizations, and administrative units. The committee was established on 3

April 2025 and operates under the direct supervision of the Rector, reporting annually to the Academic Council.

Its responsibilities include:

- Coordinating institutional sustainability initiatives and implementation of the Sustainable Development Goals.
- Monitoring performance indicators across university departments.
- Preparing annual sustainability reports and supporting external sustainability assessments.
- Coordinating awareness campaigns, student engagement, and sustainability-related research.
- Promoting transparent data management and international cooperation.

The committee includes senior university leadership and representatives responsible for academic affairs, research and innovation, finance, youth affairs, environmental management, gender and inclusion, anti-corruption, student leadership, and other institutional functions.

A dedicated Compliance Control Unit operates within the University to support the implementation of anti-corruption and integrity measures. The University Code of Ethics applies to leaders, academic and administrative staff, technical personnel, researchers, doctoral candidates and students, establishing common principles of ethical conduct and professional responsibility. New employees are introduced to the University's anti-corruption and ethical requirements during the onboarding process**,** helping ensure that institutional integrity standards are communicated from the beginning of employment.

KSU provides several channels through which stakeholders can raise concerns and provide feedback. Anonymous surveys are available for academic staff, students and parents, while *QarDU_AntiKor_bot* and dedicated trust hotlines provide channels for reporting complaints, suspected corruption and other institutional concerns.

Karshi State University maintains security and safety arrangements to protect students, staff, and visitors and to support a safe learning and working environment.

Key measures include:

- CCTV surveillance covering academic buildings, student dormitories, and outdoor areas.
- An emergency hotline and alarm system for reporting incidents and requesting assistance.
- Trained and certified security personnel stationed across the campus.
- Security monitoring intended to prevent incidents and support rapid responses to emergencies.

These arrangements support institutional responsibility for campus safety and the protection of the university community. The University also maintains internal

information-security procedures designed to protect confidential institutional and employee information and support responsible handling of sensitive data.

The university provides facilities intended to improve access to education and campus services for people with disabilities and individuals with special accessibility needs.

Accessibility measure	Implementation
Designated parking	Parking facilities for people with disabilities
Accessible toilets	Available in main buildings, including faculties, dormitories, the rectorate, and the library
Wheelchair ramps	Installed at main entrances and along pathways
Tactile paving	Provided along corridors and key walking routes
Braille signage	Available on doors and directional signs
Accessibility lifts	Wheelchair and platform lifts in specified multistorey buildings

These facilities support independent movement, equal access to university services, and a more inclusive institutional environment.

The university's Sustainability Committee brings together senior leadership, academic departments, administrative offices, and student representatives to coordinate institutional sustainability activities. The committee's structure includes representatives responsible for finance, human resources, academic administration, gender and inclusion, anti-corruption, student leadership, communications, and environmental activities. This cross-functional structure supports coordination between university units and the monitoring of institutional objectives.

SDG 17 – Partnerships for the goals

Strengthen the means of implementation and revitalize the global partnership for sustainable development

KSU works with local, national and international partners through research, education, environmental programmes, technology initiatives and international cooperation.

- the University works with local communities and government agencies on sustainability goals.
- Eco Trans is a documented partner for waste collection and recycling.
- the University collaborates with local healthcare providers.

- curriculum modernization involves industry representatives, employers, academic experts and international partners.
- a Malaysian delegation representing Jaafar Industries Sdn Bhd and TRUCAT Global PLT visited KSU to discuss cooperation in competency-based human resource management, JAKIM halal certification, halal food production, joint projects and long-term strategic partnerships.
- international scientific and educational activities support knowledge exchange and global collaboration.

Karshi State University maintains international cooperation in education, research, innovation, and sustainable development. The university has signed agreements and memorandums of understanding with more than 70 foreign educational institutions. Research and sustainability cooperation involves universities and organizations in countries including:

- Russia
- India
- China
- Kazakhstan
- Kyrgyzstan
- Tajikistan

These partnerships support academic exchange, collaborative research, knowledge sharing, and the exchange of environmental, agricultural, and sustainability-related data. The university also participates in international conferences and initiatives that encourage cooperation on biodiversity, climate action, sustainable agriculture, ecological management, and geographic information systems.

Karshi State University collaborates with local communities, schools, youth organizations, and environmental groups to promote sustainable development.

Activities include:

- Environmental clean-up campaigns and plogging events.
- Tree-planting and green-space initiatives.
- Environmental awareness programs and educational events.
- Community engagement on water conservation and pollution prevention.
- Student volunteer initiatives involving groups such as Yosh Ekopatrul.
- Regional environmental campaigns involving academic departments and local partners.

Between 2023 and 2025, the university conducted or participated in more than 15 environmental improvement campaigns across Karshi and the Kashkadarya region. During 2024–2025, it also organized and supported more than 30 sustainability-related events.